HEATED OXYGEN SENSOR (HO2S) [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

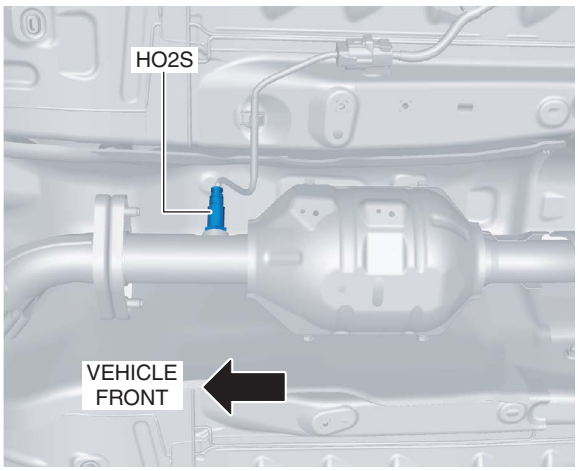
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Purpose/Function

- Detects the oxygen concentration (theoretical air/fuel ratio) in the exhaust gas as basic information (mainly used to determine the fuel injection amount and the deterioration of catalytic converter) and sends it to the PCM as an oxygen concentration (theoretical air/fuel ratio) signal.

Construction

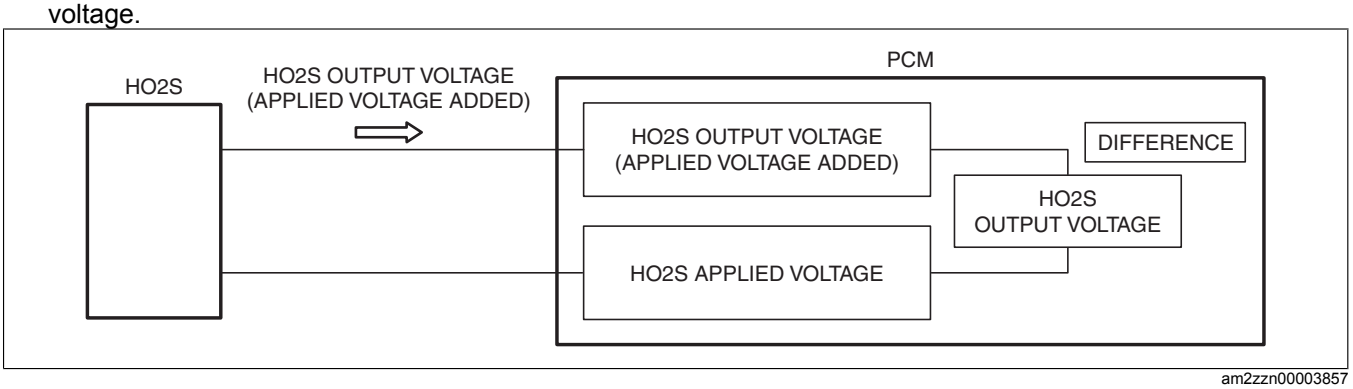
- HO2S is located on the pipe between the WU-TWC and TWC of the catalytic converter.
- A zirconia element has been adopted (as the detecting element) for detecting the oxygen concentration (theoretical air/fuel ratio).



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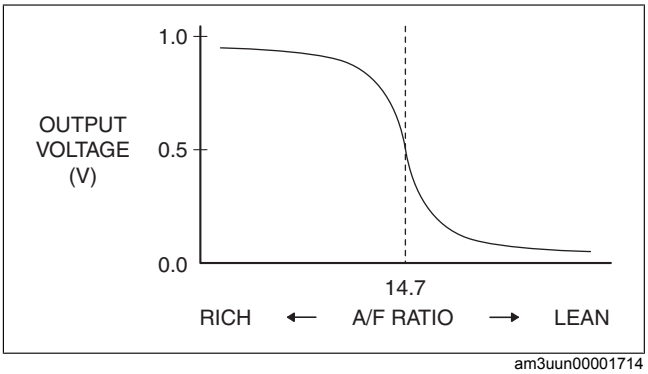
Operation

- Sends the signal, which fluctuates according to the oxygen concentration (theoretical air/fuel ratio), to the PCM.
- The PCM applies **approx. 1.5 V** to the HO2S output voltage to prevent incorrect diagnosis of the HO2S output voltage.



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HO2S characteristic



am3uun00001714

Fail-safe

DTC No.	Fail-safe function
P0037:00	• Stops fuel feedback control.
P0038:00	• Stops fuel feedback control.
P0137:00	• Fixes duty value of A/F sensor heater • Stops fuel feedback control of A/F sensor